

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

[Week 1 Activities]

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Course Title: Software Development I
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Section: 4B

Submitted to:
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Software Requirements Specification (SRS)

Project Title: Chess Game

Purpose

The purpose of this project is to develop an interactive Java-based Chess Game that provides a complete two-player chess experience through a graphical user interface. The project aims to apply Object-Oriented Programming (OOP), event handling, data structures, and logical problem-solving in a practical software application. The system will allow players to play chess according to standard rules while automatically managing piece movement, turn switching, capturing, check, checkmate, and game results.

Scope

- The Chess Game is a Java-based application designed to provide an interactive two-player chess experience.
- The system will simulate a standard chess game on a board, including piece movement, turn management, capturing, and essential chess rules.
- The project will focus on applying, GUI design, event handling, and game logic to develop a structured and user-friendly application.

System Overview

The Chess Game provides an interactive chessboard where two players can play against each other according to standard chess rules. It manages piece movement, turn switching, capturing, move validation, check, checkmate, castling, and pawn promotion. The system also provides basic game controls. The application follows an object-oriented structure to keep the game logic organized and maintainable.

Functional Requirements

- Provide an interactive chessboard.
- Implement all chess pieces: King, Queen, Rook, Bishop, Knight, and Pawn.
- Initialize pieces according to standard chess positions.
- Support legal movement for each type of chess piece.

- Prevent illegal moves.
- Support piece capturing.
- Manage alternating turns between White and Black.
- Display the current player's turn.
- Detect Check, Checkmate, and game completion.
- Implement Castling and Pawn Promotion.
- Display the winner and game result.

Non-Functional Requirements

- **Technology:** Java with Java Swing.
- **User Interface:** Simple, clear, and easy to operate.
- **Performance:** Provide quick response to player actions.
- **Reliability:** Maintain correct board state and chess rules throughout the game.
- **Maintainability:** Use modular classes and OOP principles for organized code.
- **Usability:** Allow players to understand the current game state through clear visual indicators and messages.
- **Error Handling:** Prevent invalid operations and handle unexpected user actions appropriately.

Expected Outcome

The expected outcome is a fully functional and user-friendly Chess Game that allows two players to play a complete chess match on a computer. The system will correctly handle legal and illegal moves, player turns, piece captures, check and checkmate conditions, and essential special rules. It will clearly display the current game status and announce the winner when the game ends. The project will also demonstrate practical implementation of Java, OOP, GUI development, event handling, and game logic, resulting in a reliable and presentable project.